Actual target-branch radical is the witness line

November 13, 2026. Symbolic transfer; unsigned evidence [pcc:unverified].

Objects and guards. Over any field, let F, H be monic and $0 < \deg H$, $\deg H + 1 < \deg F$, $UF + VfH = 1$. Define

$$\mu(P) = [X^{\deg F - 1}](PV \bmod F), \qquad \mathcal{B}((A,B),(Q,T)) = \mu(AQ - fBT).$$

Both polynomial pairs have degree caps $(\deg F, \deg H)$. The actual branch is $\ell(A,B) = A(q) - yB(q)$. Assume $F(q) \neq 0$, $\alpha \neq 0$ and $\ell(\alpha F, \beta H) = 0$. For every admissible branch pair (A,B):

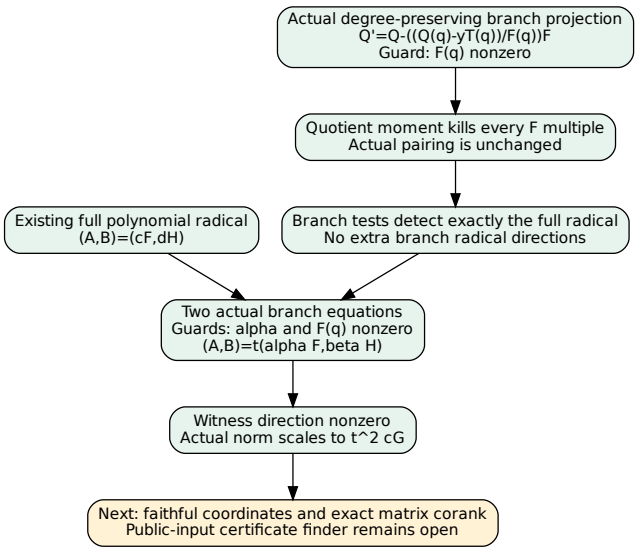
$$\mathcal{B}((A,B),(Q,T)) = 0 \text{ for all admissible branch tests} \quad \Longleftrightarrow \quad (A,B) = t(\alpha F, \beta H).$$

The witness direction is nonzero. If its supplied actual norm is cG, the norm on that line is $t^2 cG$. No literal matrix rank is assumed.

Constructive proof. Set $s = (Q(q) - yT(q))/F(q)$ and $Q' = Q - sF$. Then (Q', T) is on the branch and retains its degree caps. The exact identity

$$A(Q - sF) - fBT = (AQ - fBT) - F(sA)$$

and annihilation of every modulus multiple give the same actual pairing. Branch tests therefore detect exactly the full radical. The existing block radical theorem gives $(A,B) = (cF, dH)$. The two branch equations imply $\alpha d = \beta c$ after cancelling $F(q)$, and $t = c/\alpha$ gives the line. There is no division by 2, y, $H(q)$ or β.



Formal evidence and next transfer

Formal map. Frozen source `38b91e915`, `NormBranchRadical.lean`, has nine declarations: Its nine declarations cover the universal projection identity, actual moment annihilation, branch projection and pairing preservation, restricted versus full tests, scalar elimination, the witness-line radical, witness nonzeroness and actual norm scaling. Exact hypotheses and unsigned producer PCC bindings are linked from the Markdown report.

Checks. Target build004 completes3148 jobs with pinned Lean4.35.0, CryptoLean148f5dc and Mathlibcfc55fc. The full tracked-source gate and fresh import-inclusive replay have attributable completion receipts. Only standard foundations are admitted; no signed PCC verdict is claimed. Draft001 failed on opaque moment unfolding and a division implication. Draft002 uses the exported momentApply and a direct equality; builds003/004 add the universal identity, nonzero witness and norm scaling. Failed logs are retained.

Native record `IDC1h75cd0c4ee7db5e5d` checks only the universal CommRing projection identity at32 statement-derived plus32 extra-stream points through the unchanged maintained checker. It does not execute Lean or prove branch guards, radical dimension or search cost. Its22 inherited tests pass.

Certificate scope. The supplied witness and Bezout data determine the quotient functional. This result gives no public-input method for finding them. The annotated nonscalar-origin/jet-completion proposal has a separate, unproved reachability obligation; this functional is not shown to be reachable as $W_{\mathrm{base}}+C(T)H(T)$. The lower norm polynomial named H here does not construct that jet correction. Constant-H and degree-drop charts remain separate.

Next proof obligations. Transfer the actual product-basis coordinates faithfully to the two bounded polynomial spaces, including degree bounds and injectivity. Expand the matrix bilinear expression for arbitrary vectors, then use a genuine basis of the branch kernel. Transfer the single nonzero radical line to exact matrix corank, a nonzero adjugate and every stationarity guard required by recovery. The finder and its uniform bit cost remain additional obligations.

Sources. Prior local block-radical, branch-test and moment theorems; source-linked matrix/functional continuation plans; pinned Mathlib polynomial natDegree bounds. Primary summation-polynomial context remains Galbraith15.8.1–2, source-pinned in the cost audit. This elementary projection has no priority claim. KB MCP/wrapper and conductor unavailable; bounded fallback only.

Resources. Local proof and expression-correctness checks only. No Python mathematical research, new paid job or timing/scaling campaign. Monthly isolab budget remains exhausted. No benchmark, leaderboard or ecdb result; canonical scoreboards and historical measured figures remain unchanged. Editable DOT/SVG, source-linked report and exact receipts accompany this PDF.